12 Place Solution

Rank	12
Team	DeadKey
Author	DeadKey
Topic ID	550400
Version	Original

Source: https://www.kaggle.com/competitions/um-game-playing-strength-of-mcts-variants/discussion/550400
Retrieved with Kaggle CLI on 2026-07-03.

Hello everyone,
Here’s my short writeup on achieving a solo gold medal in the MCTS Competition.

Ensemble

My final solution was an ensemble of my unique approach combined with the public notebooks MCTS Starter and MCTS: DeepTables NN. A significant missed opportunity was that I didn’t leverage any data augmentation, which could have further improved the results.

Unique Solution

The core idea behind my unique solution was to compute a better approximation of the most likely outcome between two random MCTS agents—essentially, an improvement over the AdvantageP1 feature.

To create this new feature, I trained a DeBERTa model to predict the most likely game outcome based on the content of the LudRules column. Additionally, I trained a LightGBM model to further refine the predictions.

You can find the notebook showcasing the unique part of my solution here:https://www.kaggle.com/code/dettki/mcts-private-rank-12-unique-part

Supplemental Q&A

> **Gabor Balazs · 2024-12-07T15:45:09.730000**
>
> Thank you for sharing your solution and congratulations for your solo gold medal. I think the definition of `cb_params` is missing from the linked version of the notebook, so we cannot see what CatBoost parameters you used. Would you mind adding that?

> **DeadKey · 2024-12-07T16:21:19.510000**
>
> Thanks. i used this cb_params:
> cb_params = {'grow_policy': 'SymmetricTree', 'n_estimators': 2000, 'learning_rate': 0.08617153230342124,
> 'l2_leaf_reg': 1.0036132233587023,
> 'max_depth': 10, 'random_strength': 0.3,
> 'verbose': 500,"task_type": "GPU"}
> cb_params={'task_type' : "GPU",
> 'eval_metric' : "RMSE",
> 'bagging_temperature' : 0.60,
> 'iterations' : 3096,
> 'learning_rate' : 0.08,
> 'max_depth' : 10,
> 'l2_leaf_reg' : 0.25,
> 'min_data_in_leaf' : 24,
> 'random_strength' : 0.20,
> 'max_bin' : 2048,
> 'verbose' : 0,

> }
> They are close to yunsuxiaozi in his great notebook, I tried to tune them a bit but didnt seem to me that there was much to gain there.

> **DeadKey · 2024-12-07T16:30:15.503000**
>
> If you are looking for the things that had the most impact. I would they that it was the oof_predictions i already mentioned. The label_encoding of the agent columns and the stratification by the agent-pairing during training.